

A function

A function assigns exactly one output to each input in its domain.

Can two different inputs have the same output?

A relation from ordered pairs

$\{(-1, 4), (0, 4), (2, 7)\}$

Does this relation represent a function?

A repeated input

$\{(1, 2), (1, 5), (3, 6)\}$

Does this relation represent a function?

Function notation

The notation $f(x)$ names the output of f at input x .

$$f(3) = 8$$

What ordered pair belongs to the graph of f ?

Evaluate at a number

$$f(x) = x^2 - 3x + 1$$

What is $f(-2)$?

Evaluate at an expression

$$f(x) = 2x^2 - 5$$

What is $f(a + 1)$?

Find an input

$$f(x) = x^2 + 1$$

Which inputs make $f(x) = 10$?

Evaluate a model

$C(n)$ is the total cost in dollars of n tickets.

$$C(n) = 12n + 5$$

What does $C(4)$ represent?

One-to-one functions

A one-to-one function assigns different outputs to different inputs.

Why is this stronger than being a function?

Test one-to-one behavior

$$f(x) = x^2$$

Is this function one-to-one on all real numbers?

The vertical line test

A graph represents y as a function of x exactly when every vertical line meets it at most once.

$$x^2 + y^2 = 9$$

Does this equation define y as a function of x ?

The horizontal line test

A function is one-to-one exactly when every horizontal line meets its graph at most once.

$$f(x) = 2x - 3$$

Is this function one-to-one?

A constant function

$$f(x) = 3$$

What does its graph look like?

The identity function

$$f(x) = x$$

What does its graph look like?

A linear function

$$f(x) = 2x + 1$$

What features locate its graph?

The quadratic function

$$f(x) = x^2$$

What is its basic graph shape?

The cubic function

$$f(x) = x^3$$

How does its graph differ from the quadratic graph?

The absolute value function

$$f(x) = |x|$$

What is its basic graph shape?

The square root function

Use real inputs and outputs.

$$f(x) = \sqrt{x}$$

Where does its graph start?

The cube root function

Use real inputs and outputs.

$$f(x) = \sqrt[3]{x}$$

Can its graph include negative input values?

The reciprocal function

$$f(x) = \frac{1}{x}$$

Where are its two branches located?

The reciprocal squared function

$$f(x) = \frac{1}{x^2}$$

How does its graph differ from the reciprocal graph?

A difference of outputs

$$f(x) = x^2 + 2x$$

What is $f(3) - f(1)$?

A difference quotient

Use $h \neq 0$ and $f(x) = x^2$.

$$\frac{f(x + h) - f(x)}{h}$$

How can we simplify this quotient?